

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn import metrics
import seaborn as sn
import matplotlib.pyplot as plt
```

```
from google.colab import files
uploaded = files.upload()
```

No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving binary.dta to binary.dta

```
data = pd.read_stata('binary.dta')
print(data.shape)
data.head()
```

```
(400, 4)
   admit  gre  gpa  rank
0    0.0 380.0 3.61   3.0
1    1.0 660.0 3.67   3.0
2    1.0 800.0 4.00   1.0
3    1.0 640.0 3.19   4.0
4    0.0 520.0 2.93   4.0
```

```
data.isnull().sum()
```

```

      0
admit 0
gre    0
gpa    0
rank   0

dtype: int64
```

```
feature_cols = ['gre', 'gpa', 'rank']
x = data[feature_cols]
y = data['admit']
```

```
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=5)
display(x_train.shape, y_train.shape, x_test.shape, y_test.shape)
```

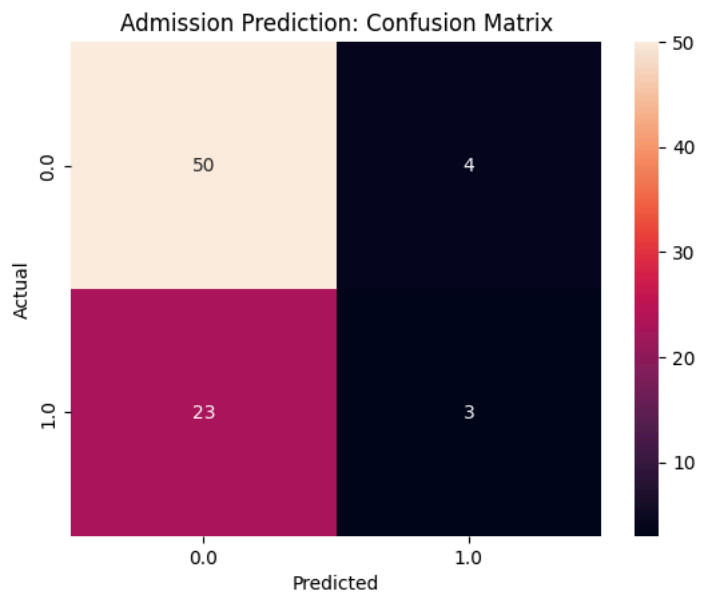
```
(320, 3)
(320,)
(80, 3)
(80,)
```

```
model = LogisticRegression(solver='lbfgs', max_iter=1000)
model.fit(x_train, y_train)
y_pred = model.predict(x_test)
```

```
conf_mat = metrics.confusion_matrix(y_test, y_pred)
print('Confusion Matrix : ', conf_mat)
Accuracy_score = metrics.accuracy_score(y_test, y_pred)
print('Accuracy Score : ', Accuracy_score)
print('Accuracy in Percentage : ', int(Accuracy_score*100), '%')
```

```
Confusion Matrix : [[50  4]
 [23  3]]
Accuracy Score : 0.6625
Accuracy in Percentage : 66 %
```

```
conf_mat = pd.crosstab(y_test, y_pred, rownames=['Actual'], colnames=['Predicted'])
sn.heatmap(conf_mat, annot=True, fmt='d')
plt.title('Admission Prediction: Confusion Matrix')
plt.show()
```



Start coding or [generate](#) with AI.